

OMAR REHAN

AI / MACHINE LEARNING / COMPUTER VISION ENGINEER

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#) | [Hugging Face](#) | [Kaggle](#) | [Medium](#) | [Tableau](#)

PROFESSIONAL SUMMARY

Results-driven AI, Machine Learning, and Computer Vision Engineer with a B.Sc. in Computer Science and over 3 years of hands-on experience designing, training, and deploying end-to-end deep learning, 3D medical vision, audio classification, and MLOps systems. Proficient in PyTorch, MONAI, TensorFlow, Keras, Hugging Face, and Python, with a track record of implementing optimized inference pipelines and MLOps workflows. Skilled in fine-tuning vision-language models (Llama 3.2-Vision, Qwen2.5-VL) using Unsloth/LoRA, implementing YOLOv8m-OBb object detection, training 3D SegResNet segmentation and Mel-spectrogram audio CNNs, and integrating Grad-CAM interpretability. Experienced in containerizing AI applications with Docker and deploying REST APIs to Hugging Face Spaces and AWS.

EDUCATION

Arab Academy for Science, Technology & Maritime Transport (AASTMT)

Oct 2019 - Feb 2024

B.Sc. in Computer Science · Graduation Project: AI-Powered Smart Classroom Automation, TensorFlow Lite CNN on Raspberry Pi; Flutter/Bluetooth IoT control of lighting, HVAC & projectors.

PROFESSIONAL EXPERIENCE

Freelance AI & Machine Learning Engineer

2022 - Present

- Consulted for global clients to design and deploy custom computer vision pipelines, deep learning classifiers, and vision-language systems using **PyTorch**, **TensorFlow**, and **Hugging Face**.
- Developed and containerized REST APIs (**FastAPI**, **Docker**) and interactive **Gradio** dashboards, deploying optimized inference endpoints to **AWS** and Hugging Face Spaces.

IT Support & Data Management Specialist

Apr 2024 - May 2025

Managed digital records, database maintenance, and network infrastructure for the Egyptian Army (Electronic Warfare Dept, North Sinai); contributed to software development, script automation, and AI integration initiatives.

SELECTED TECHNICAL PROJECTS

MEDTRACE: Longitudinal AI for Brain-MRI Disease Evolution

2026

- Trained a 3D **MONAI SegResNet** (18.8M params) on BraTS 2023 GLI across 4 co-registered sequences (T1c, T1n, T2-FLAIR, T2w) to segment 3 overlapping compartments (WT, TC, ET), achieving **0.894 mean test Dice** (0.922 WT, 0.908 TC, 0.852 ET).
- Engineered an end-to-end longitudinal pipeline for 209 patients (874 studies) with **SimpleITK** registration and volumetric tracking; built a full-stack workstation with 2D viewer (**Cornerstone3D**), 59.9 FPS 3D ray-casting (**vtk.js**), and evidence API (**FastAPI**, **Next.js**, **Docker**).

SIVED: SAR Satellite Vehicle Detection

2026

- Engineered an oriented object detection pipeline utilizing **YOLOv8m-OBb** on 1,044 SAR satellite images to identify vehicles at arbitrary orientations, achieving **98.8% mAP@0.5** and 81.6% mAP@0.5:0.95.
- Optimized training by disabling color augmentations for grayscale radar inputs and scaling HSV value (0.4) for backscatter variations; deployed as a **Gradio** web app on Hugging Face Spaces.

AstroVision: Multimodal VLM Fine-Tuning

2025

- Fine-tuned Meta Llama 3.2-11B Vision and Qwen2.5-VL-7B for domain visual reasoning using **Unsloth**, applying 4-bit quantization and **LoRA** (r=16, alpha=16) to cap peak memory to 10 GB VRAM.
- Structured a custom dataset of 250 astronomy image-caption pairs into conversation-style formats, evaluating generation via ROUGE-L (0.24) and BLEU, and deployed inference via **Gradio**.

Deep Audio Classifier: Mel-Spectrogram CNN Pipeline

2025

- Engineered an audio classification pipeline converting raw audio to **128×128 Mel spectrograms** via **Librosa** (16 kHz resampling, silence trimming, peak normalization, pitch/time augmentations).
- Designed and trained a deep **TensorFlow/Keras CNN** with Conv2D, BatchNorm, and Dropout; implemented SHA-256 hash deduplication, automated LR decay, and early stopping; deployed demo to **Hugging Face Spaces**.

ResNet50V2 Chest X-Ray COVID-19 Classification & Grad-CAM

2025

- Trained a **ResNet50V2** transfer learning classifier; accelerated **Grad-CAM** latency by **90%** (16s to 1s) via layer caching and **@tf.function** compilation; containerized with **Docker** and deployed to Gradio.
- Additional AI, computer vision, and deep learning projects are available on the [Portfolio](#).**

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, SQL, HTML/CSS

AI & ML: Deep Learning, Machine Learning, MONAI, Audio Processing, NLP, Transfer Learning, Fine-Tuning, PEFT, LoRA, Quantization, Scikit-learn, LangChain, Unsloth, Transformers

Computer Vision & Signal Processing: 3D Segmentation, SegResNet, YOLOv8-OBb, SimpleITK, vtk.js, Cornerstone3D, Mel-Spectrograms, Librosa, OpenCV, Image Classification, Object Detection, Grad-CAM

Frameworks: PyTorch, TensorFlow, Keras, Next.js, React, FastAPI, Flask, Streamlit, timm

Deployment & MLOps: Docker, Docker Compose, Hugging Face Spaces, Gradio, AWS, Vercel, REST APIs, Model Optimization, Inference

Databases: PostgreSQL, MongoDB, MS SQL Server, Firebase, Supabase

Cloud & Developer Tools: AWS, Google Cloud Platform (GCP) | Git, GitHub, VS Code, Google Colab, Linux, Raspberry Pi

CERTIFICATIONS

AI Associate & IoT Associate, Digital Hub (IBM-based)

Jan 2023

CCNAv7: Enterprise Networking, Security & Automation · Switching, Routing & Wireless · Intro to Networks

2022 - 2023

Soft Skills, HRDC Cairo

Aug 2021

COMPETITIONS

Hult Prize, Regional Finals Competitor, Cairo Impact Summit (UN-affiliated)

Apr 2021

LANGUAGES

Arabic: Native | English: Full Professional Proficiency